

Accurate and Efficient Phylogenetic Inference through End-To-End Deep Learning

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Abstract

Accurate phylogenetic inference is crucial for understanding evolutionary relationships among species. Deep learning technique has been introduced for phylogenetic inference; however, the existing deep learning-based approaches either suffer from limited accuracy as they split inference into several disjoint stages, or exhibit low efficiency and hardly apply to the cases with over 20 species. Here, we present an accurate and efficient approach to phylogenetic inference. Our approach, called NeuralNJ, employs an end-to-end framework that directly constructs phylogenetic trees from the input taxa, thus effectively avoiding the inaccuracy incurred by the split inference stages. The key innovation of NeuralNJ lies in its learnable neighbor joining mechanism, which iteratively joins neighbors guided by learned priority scores and thereby achieves accurate tree reconstruction. The inference accuracy is further enhanced through incorporating reinforcement learning-based tree search. Using both simulated and empirical data as representatives, we demonstrate that NeuralNJ can effectively infer phylogenetic tree with improved computational efficiency and reconstruction accuracy. The study paves the way to accurate and efficient phylogenetic inference for hundreds of taxa in complex evolutionary scenarios.

Keywords: phylogenetic inference, phylogenetic tree, deep learning

Introduction

Phylogenetic inference aims to reconstruct evolutionary relationship among organisms through analyzing their genetic information, especially molecular sequences (Brocchieri 2001). This analytical approach has become fundamental to a variety of biological tasks, including systematics (Rannala and Yang 2008), molecular evolution (Lythgoe et al. 2021; Shao et al. 2021; Hoyt et al. 2022), and molecular ecology (Gómez-Corralles and Prada 2020; Pickering et al. 2022). For example, phylogenetic analysis of COVID-19 strains provided invaluable insight into their evolutionary history, thus greatly facilitating timely evidence-based public health decisions (Morel et al. 2020; Li et al. 2021). Despite its great importance, phylogenetic inference is challenged as the number of possible tree topologies grows super-exponentially with the number of species (Felsenstein 1978; Chor and Tuller 2005). Thus, developing accurate and efficient phylogenetic inference approaches is highly desired.

Classical phylogenetic inference approaches, which rely on expert-designed heuristics to construct phylogenetic trees, can be divided into two categories, namely, distance-based and character-based approaches. The distance-based approaches first estimate pairwise distances among the given species, and then construct a phylogenetic tree that optimally satisfies the estimated distance constraints through clustering or neighbor-joining (NJ) (Sokal and Michener 1958; Fitch and Margoliash 1967; Jukes and Cantor 1969; Saitou and Nei 1987). Unlike the distance-based approaches using a single number to summarize the difference between two species, the character-based approaches explore full information of

organisms' characters (traits, features, or DNA sequences) to find the most plausible phylogenetic tree explaining the character distribution. The widely-used explaining principles include maximum parsimony, which builds a tree with the minimized character changes (Farris 1970; Swofford and Sullivan 2009), maximum likelihood (ML) under a specific evolutionary model (Felsenstein 1981; Guindon et al. 2010; Nguyen et al. 2014; Kozlov et al. 2019; Minh et al. 2020), and Bayesian inference that identifies the tree with the maximum posterior probability (Felsenstein 2004; Price et al. 2010; Ronquist et al. 2012; Bouckaert et al. 2014). The phylogenetic tree can be constructed either through tree building (such as NJ) or through iterative improving a tree using rearrangement operations like subtree pruning and regrafting (SPR) (Bordewich and Semple 2005) and nearest neighbor interchange (NNI) (Waterman and Smith 1978; Desper and Gascuel 2002).

Recently, machine learning, especially deep learning, has emerged as a promising paradigm for phylogenetic inference. These approaches learn key elements of phylogenetic inference directly from training data, specifically elements such as pairwise distance (Nesterenko et al. 2025), tree rearrangement strategies (Azouri et al. 2021, 2024), and tree likelihood estimation (Xie and Zhang 2023; Zhang 2023). Generative adversarial network has also been incorporated into phylogenetic inference, which attempts to yield a phylogenetic tree that generates multiple sequence alignment (MSA) indistinguishable from the input MSA (Smith and Hahn 2023).

The machine learning-based approaches, although exhibiting promising performance, still suffer from several

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limitations: (i) Some approaches, such as PhyDL (Zhou et al. 2023) and PhyloGame (Azouri et al. 2024), focus on quartet molecular phylogeny (Ranwez and Gascuel 2001) or small datasets with at most 20 taxa, thus restricting themselves from larger phylogenetic inference tasks. (ii) Other approaches, such as VBPI-GNN (Zhang 2023), ARTree (Xie and Zhang 2023), focus on the accuracy of posterior likelihood of phylogenetic trees but pay little attention to the efficiency of tree construction. (iii) Only partial modules of the inference procedure are trained, e.g. Phyloformer (Nesterenko et al. 2025) trains the distance estimating module only and simply uses the traditional distance-based tree construction module, thus making the training errors cannot be sufficiently explored to optimize all modules.

Here, we present an accurate and efficient approach (called NeuralNJ) to phylogenetic inference which constructs the tree directly from genome sequences. Our approach adopts an end-to-end framework that propagates the training loss from the final layer back to the input layer, thus enabling optimizing all intermediate modules along the propagation and improving accuracy of phylogenetic inference. In addition, rather than learning pairwise distances between species, which is dependent on accurate distance estimation and sensitive to branch length variations, our approach learns to rank candidate subtrees using a learning-to-rank strategy, potentially offering advantages in both phylogenetic inference stability and accuracy. Our approach constructs phylogenetic trees in just one pass rather than multiple iterative steps for refinements, thus gaining the advantages of high efficiency. We evaluated NeuralNJ on simulated and empirical datasets to highlight its special features in both accuracy and efficiency. The source code of NeuralNJ, together with the datasets for training and evaluation, are publicly available at <https://github.com/ZhangXinru99/NeuralNJ>.

Results

Overview of NeuralNJ Algorithm

We describe the main steps and operations of NeuralNJ in Fig. 1. NeuralNJ takes genome sequences in the form of MSA as its input and constructs a phylogenetic tree using an encoder-decoder architecture that consists of a sequence encoder and a tree decoder. The details of these two modules are described as follows:

- Sequence encoder:** Sequence encoder embeds each input sequence into a high-dimension vector to capture its essential characteristics. For this aim, we implement sequence encoder using the MSA-transformer architecture (Rao et al. 2021). The advantage of this architecture is that it alternately computes attention along both species and sequence dimensions of the input MSA, and thus generates site-aware and species-aware representations (see Supplementary Algorithm 2 for further details). These representations, rather than the original genome sequences, are explored by the subsequent tree decoder for constructing phylogenetic trees.
- Tree decoder:** Tree decoder aims to decode a phylogenetic tree from the representations of species calculated by sequence encoder. Tree decoder starts from an initial subtree set that consists of each species as a degenerated tree with only one node, and then iteratively selects and joins two subtrees until acquiring a complete phylogenetic tree (Fig. 1b). To select the most plausible subtrees for

joining, we enumerate all possible subtree pairs, estimate the embedding of the parent node of each subtree pair, and exploit this embedding to calculate the priority score that the two subtrees of this pair, if joined together, appear in the ground-truth phylogenetic tree.

It should be noted that, when estimating the embedding of the parent node of two subtrees, we consider not only the two subtrees themselves but also all other subtrees using a topology-aware gated network (Fig. 1c). This design is motivated by the key insight that a newly-added parent node (say h_x in Fig. 1c) serves as a topological bridge between two subtrees (h_l and h_r) and other subtrees that might be merged in future steps (h_{rest}). To accurately capture this bridging role, NeuralNJ employs a dual-level fusion strategy: local topology is encoded through gated left-right subtree fusion, while global ancestry information is incorporated through attention-based aggregation of remaining subtrees (see further details in Supplementary Algorithm 3).

We have also implemented three variants of NeuralNJ, including: (i) NeuralNJ that greedily selects the subtree pair with the highest score, (ii) NeuralNJ-MC that samples from all subtree pairs according to their scores, and (iii) NeuralNJ-RL that uses reinforcement learning strategy with likelihood as reward rather than supervised learning. The latter two variants usually generate multiple complete trees, from which we select the highest likelihood one. Here, the likelihood of a complete phylogenetic tree is calculated using Felsenstein's pruning algorithm (Felsenstein 1981), which efficiently computes the likelihood by a post-order traversal of the tree.

We trained NeuralNJ on simulated data that were generated using a similar approach as Phyloformer. NeuralNJ works in an end-to-end manner, in which the training loss (the difference between a predicted tree and its ground-truth counterpart) is propagated back to the first layer. This way, both sequence encoder and tree decoder can be improved during the training process, thus facilitating learning and promoting phylogenetic inference.

NeuralNJ, once trained, can be executed directly to infer phylogenetic tree for a collection of genome sequences. We use a dataset containing 20 taxa as an example to demonstrate the phylogenetic inference process. As shown in Fig. 1d, taxon 10 and 5 were selected to join together, and the joined tree was further selected to join with taxon 16. At step 8–14, the subtree consisting of taxon 10, 5, and 16 was joined with taxon 3 and 11. At step 15–20, the subtree at the bottom was joined with the subtrees on top, thus forming a complete phylogenetic tree.

NeuralNJ Achieves High Accuracy on Simulated Data

We first evaluated NeuralNJ on simulated data that were generated using the well-established evolutionary model GTR+I+G (Tavaré 1986; Yang 1994). Each dataset consists of: (i) an evolutionary tree with a random topology and branch length sampled according to an exponential distribution, (ii) a number of taxa with sequences generated according to the evolutionary tree (see Materials and Methods for details). The evolutionary trees were used as ground-truth in subsequent evaluation procedure.

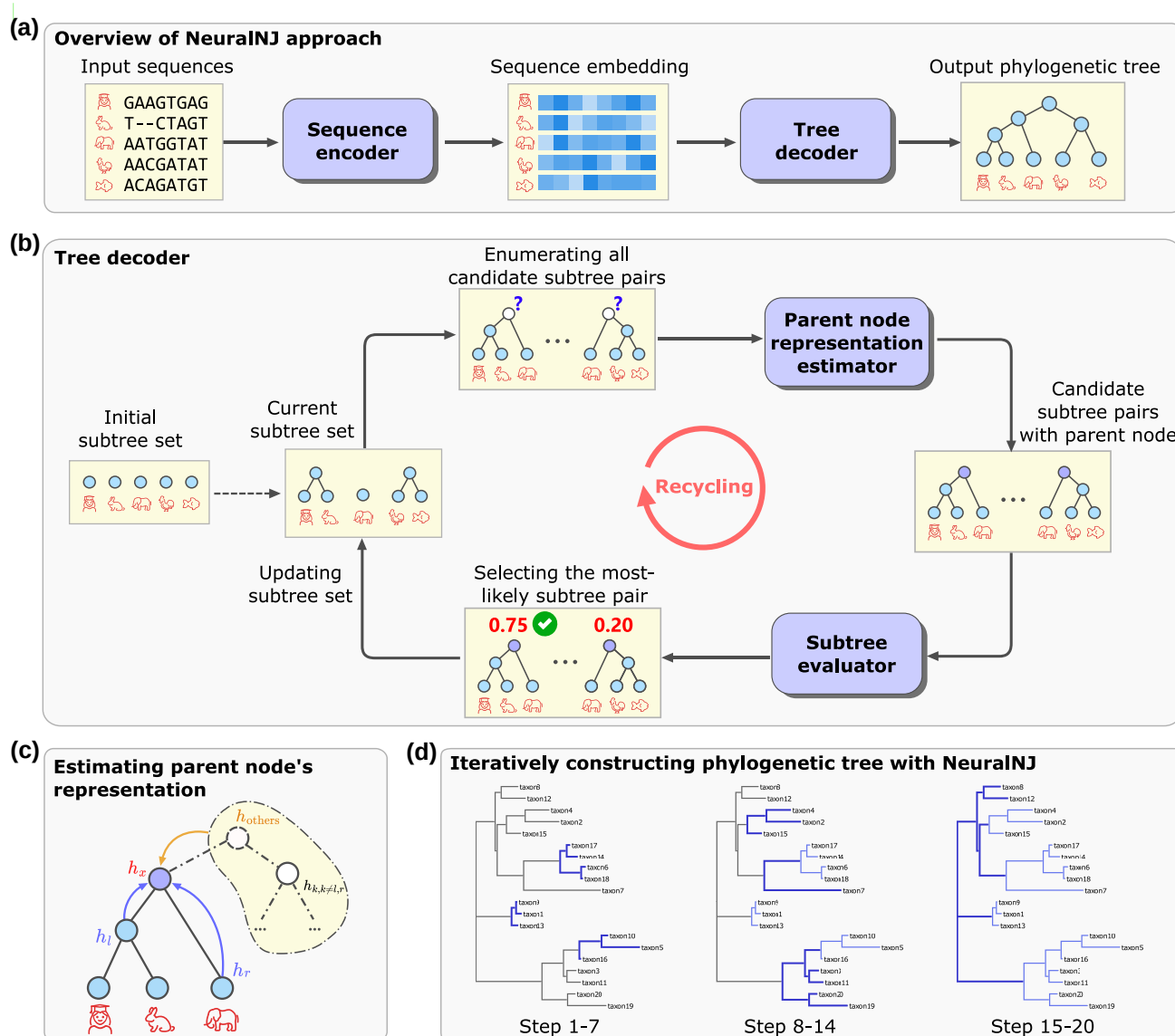


Fig. 1. Overview of NeuralNJ architecture and workflow. a) The overall architecture of NeuralNJ, consisting of a sequence encoder that transforms input sequences into embeddings and a tree decoder that constructs the phylogenetic tree. b) Detailed illustration of the tree decoder's iterative process, showing how subtree pairs are evaluated and selected. c) Mechanism for estimating parent node representation through topology-aware fusion of local and global information. d) Example of iterative phylogenetic tree construction using NeuralNJ, demonstrated in three stages (Steps 1–7, 8–14, and 15–20).

We constructed a simulated training dataset of 50-taxon phylogenetic tree-MSA pairs. For each of 29 sequence lengths (128–1,024 nt in 32-nt steps), 1,000 distinct tree topologies with corresponding MSAs were generated, totaling 29,000 training samples. Generally, shorter sequences represent more challenging tasks as they provide less information relative to longer sequences. Similarly, we constructed a test dataset comprising 640 samples. To evaluate the generalization ability of NeuralNJ, we set the test dataset with diverse taxa number (20, 50, 100 taxa) and sequence lengths (256, 512 nt). It should be noted that the training and test datasets were independently generated to avoid potential overlap.

We compared NeuralNJ with two state-of-the-art approaches: RAxML-NG, a well-established ML-based solver, and Phyloformer, a machine learning-based approach. We evaluated the constructed phylogenetic trees using the normalized RF distance (Robinson and Foulds 1981), which quantifies the topological difference between a predicted tree and its ground-truth counterpart.

Figure 2 presents the topological accuracy of NeuralNJ, Phyloformer, and RAxML-NG across data configurations: the left panel shows performance with fixed sequence length (1,024 nt), while the right panel shows performance with fixed taxa number (100 taxa). As shown in the left panel, RAxML-NG (blue bar) and Phyloformer (purple bar) achieve mean normalized RF distances of 0.108 and 0.116, respectively, on the dataset with 20 taxa, while NeuralNJ (red bar) and its variants, NeuralNJ-MC (orange bar) and NeuralNJ-RL (yellow bar), achieve distances of 0.114, 0.089, and 0.087, respectively. The right panel shows that for sequences of length 256 nt, RAxML-NG achieves a normalized RF distance of 0.316, Phyloformer achieves 0.300, while NeuralNJ, NeuralNJ-MC, and NeuralNJ-RL achieve distances of 0.269, 0.268, and 0.262, respectively. These results demonstrate the higher accuracy of our NeuralNJ approach over RAxML-NG and Phyloformer. The performance advantage of NeuralNJ becomes more pronounced in challenging scenarios with larger numbers of taxa (e.g. 100) or shorter sequence lengths (e.g. 256 nt).

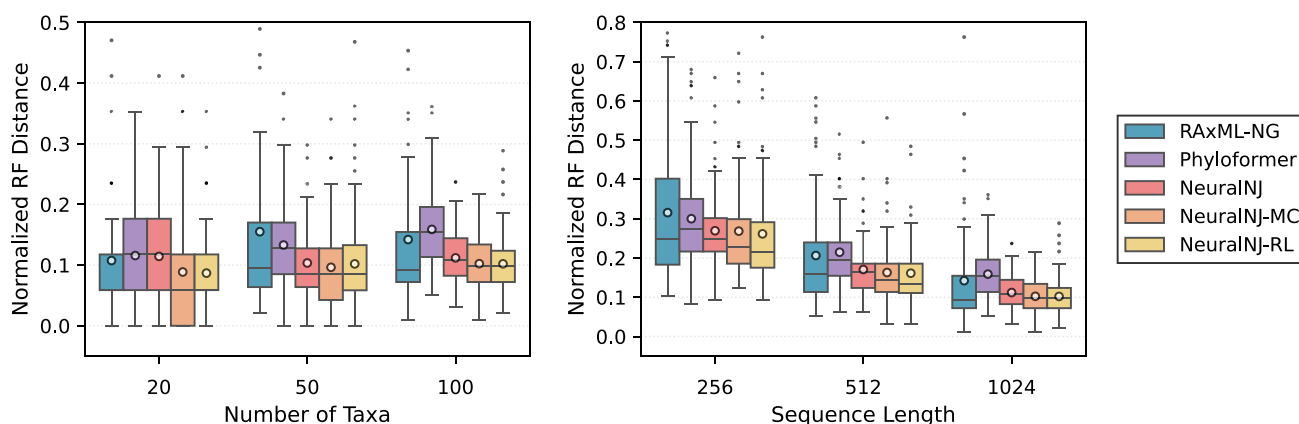


Fig. 2. Performance comparison of phylogenetic inference methods on the simulated test dataset under the GTR+I+G evolutionary model. Normalized Robinson–Foulds (RF) distances between predicted and ground-truth simulated trees across varying numbers of taxa (left) and sequence lengths (right). Lower RF distances indicate higher topological accuracy. Box plots show the median (central line), quartiles (box), 1.5 $\times$ interquartile range (whiskers), outliers (individual points), and the mean value (circle) for each group.

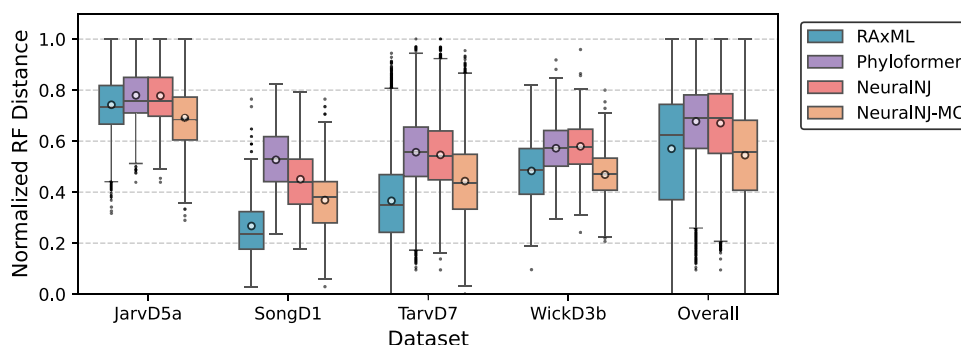


Fig. 3. Performance evaluation on empirical phylogenomic datasets. Box plots show the distribution of normalized RF distances between inferred and reference phylogenies for different inference methods across four empirical datasets (JarvD5a, SongD1, TarvD7, and WickD3b) and their overall performance. Box plots show the median (central line), quartiles (box), 1.5 $\times$ interquartile range (whiskers), outliers (individual points), and the mean value (circle) for each group, with lower RF distances indicating higher accuracy in topology reconstruction.

In summary, these results highlight the **superior performance** of NeuralNJ in **topological accuracy**, particularly for the challenging cases with larger numbers of taxa or shorter sequence lengths. Further experiments in the **Supplementary Materials** demonstrate **NeuralNJ's robust performance** across **diverse simulation scenarios**, including datasets with **longer sequences** and **different evolutionary parameters**. These evaluations consistently validate NeuralNJ's advantages in phylogenetic tree reconstruction.

NeuralNJ Achieves High Efficiency on Empirical Data

To validate our method's accuracy and efficiency on real data, we evaluated it using **four empirical phylogenomic DNA datasets**. These datasets represent diverse taxonomic groups (Zhou et al. 2017): WickD3b with 103 land plant species, JarvD5a with 48 bird species, SongD1 with 37 mammal species, and TarvD7 with 36 mammal species. Each dataset consists of multiple gene alignment sequences from these species. Both NeuralNJ and Phyloformer, trained on the **mentioned GTR+I+G simulated data**, were evaluated on these empirical datasets with varying sequence lengths based on hardware constraints: sequences up to 17,000 nucleotides for datasets with <50 taxa (achieving >98% coverage for JarvD5a, SongD1, and TarvD7) and up to 4,400 nucleotides for

WickD3b (50% coverage). Results for NeuralNJ-RL are not included in the **current empirical evaluation** because **this variant requires increased computational and hardware resources**.

Due to the absence of **true phylogenetic trees** for empirical data, we followed the **common practice** of using **widely accepted reference trees** supported by established **classical methods** (Zou et al. 2019). Following **Phyloformer's evaluation protocol** (Nesterenko et al. 2025), we used **reference trees inferred by ML-based solvers on concatenated multiple sequence alignments as ground truth** (Zhou et al. 2017). These reference trees may **inherently favor ML-based methods**, **potentially making it difficult** for our approach to **demonstrate superior performance** on such data.

We present our accuracy results in Fig. 3. As expected, **RAxML demonstrates stronger alignment** with the reference trees. On JarvD5a, NeuralNJ achieves a mean RF distance of 0.78, approaching RAxML's performance of 0.74. For SongD1, TarvD7, and WickD3b datasets, NeuralNJ shows competitive performance with mean RF distances of 0.45, 0.55, and 0.58, respectively, which are comparable to RAxML's results. NeuralNJ outperforms Phyloformer on three datasets (JarvD5a, SongD1, and TarvD7), while showing slightly higher RF distances on WickD3b. NeuralNJ-MC, which **incorporates likelihood-guided sampling strategies**, further improves upon the performance of NeuralNJ. Notably, on the JarvD5a and WickD3b datasets,

Table 1 Average runtime comparison (seconds) of different phylogenetic inference methods on empirical datasets

Dataset	RAxML ^a	Phyloformer	NeuralNJ	NeuralNJ-MC
JarvD5a	203.06	0.95	0.97	133.30
SongD1	148.82	1.01	1.37	177.40
TarvD7	99.12	0.74	0.97	151.34
WickD3b	184.26	0.95	0.91	105.26

^aRAxML timings were obtained from [Supplementary Table S6](#) of the empirical data ([Zhou et al. 2017](#)), which reports runtimes on single-gene alignments.

NeuralNJ-MC achieves lower RF distances than RAxML. Overall, NeuralNJ-MC attains comparable or marginally better accuracy, with a mean RF distance of approximately 0.53 versus 0.55 for RAxML, highlighting the potential effectiveness of our sampling-based approach. Detailed quantitative results for both the complete empirical datasets and the long-sequence subsets are provided in the [Supplementary Materials](#).

We present our efficiency results in [Table 1](#). Both NeuralNJ and Phyloformer demonstrate superior inference speed, with **runtime reductions of 100-200x compared to RAxML**. This advantage is particularly pronounced on WickD3b, the dataset with the largest number of species, where the average inference time decreases from 184.26 seconds (RAxML) to below 1 second (NeuralNJ and Phyloformer). While NeuralNJ-MC **achieve improved accuracy**, this comes at the **cost of increased computational time**, as evidenced by their long runtimes across all datasets.

In summary, NeuralNJ demonstrates **high computational efficiency** while maintaining comparable accuracy to state-of-the-art methods on empirical datasets.

NeuralNJ Perfectly Balances Accuracy and Efficiency

Both computational efficiency and inference accuracy are critical considerations for phylogenetic inference methods, particularly given the increasing scale of modern phylogenetic analyses involving large numbers of taxa and long sequences. We evaluated the trade-off between these two factors across varying problem scales under the GTR+I+G model. Our evaluation **compares NeuralNJ variants and neural baselines implemented on GPU platforms** against **classical solvers** on CPUs (detailed hardware configurations are provided in [Supplementary Table 12](#)).

[Figure 4](#) presents an evaluation using Pareto frontiers, which represent the optimal trade-off curve where no method can improve one metric without degrading another. The results are shown across varying numbers of taxa ([Fig. 4a–c](#)) and sequence lengths ([Fig. 4d–f](#)). Our analysis demonstrates that NeuralNJ consistently achieve Pareto-optimal frontiers across all experimental settings. This is particularly evident in [Fig. 4c](#), which represents a challenging scenario with 100 taxa and 1,024 nt sequences, where NeuralNJ and its variants lie on the Pareto-optimal frontier. In contrast, Phyloformer falls off the frontier with a second-order runtime and an RF distance of approximately 0.16, while RAxML-NG is suboptimal with minute-scale computational time and an RF distance of approximately 0.14. While BIONJ also maintains a position on the Pareto frontier, it demonstrates substantially lower accuracy with an RF distance of approximately 0.4 under the challenging conditions of 100 taxa and 1,024 nt sequences.

The scalability analysis reveals that NeuralNJ maintains its Pareto-optimal performance across increasing problem scales. Specifically, as the number of taxa increases from 20 to 100 ([Fig. 4a–c](#)), NeuralNJ and its variants consistently define the efficiency-accuracy frontier, while other methods deviate from optimal performance. Similarly, across increasing sequence lengths from 256 to 1,024 nt ([Fig. 4d–f](#)), NeuralNJ demonstrate robust accuracy while maintaining their computational efficiency advantage. Detailed performance analyses across different problem scales are provided in [Supplementary Section D.5](#).

While NeuralNJ and its variants (NeuralNJ-MC/RL) consistently achieve Pareto-optimal performance, they occupy different strategic positions on the frontier. NeuralNJ establishes an optimal balance point, delivering significant accuracy improvements over traditional distance-based methods while maintaining computational efficiency. In contrast, NeuralNJ-MC and NeuralNJ-RL extend toward the high-accuracy region of the frontier, offering viable alternatives for applications where precision takes precedence over computational cost.

This evaluation demonstrates that NeuralNJ achieves an optimal balance between accuracy and efficiency across diverse problem scales, while its variants provide flexible options for different accuracy-efficiency trade-offs in phylogenetic inference tasks.

The Advantages of NeuralNJ: Case Studies

[Figure 5](#) presents a case study using a simulated MSA data (50 taxa, 1,024 nt) under GTR+I+G model to visualize the reconstruction accuracy of different methods. The MSA data and corresponding true tree topology used in this case study are provided in [Supplementary Section E](#). The figure compares the true phylogenetic tree (a) with trees inferred by five methods (b–f), where red branches indicate topological inconsistencies with the true tree—specifically, join operations that do not match any valid subtree combinations in the true topology.

As illustrated in [Fig. 5](#), RAxML-NG erroneously clusters taxon39 with taxon9 and taxon50, while Phyloformer misplaces taxon50 by merging it with taxon17; in the reference tree, taxon50 should be grouped with taxon9. NeuralNJ and its variants correctly resolve the relationships involving taxon39 and taxon50, but some errors remain: NeuralNJ incorrectly merges taxon2 with the cluster containing taxon34, taxon29, taxon36, and taxon35. NeuralNJ-MC corrects this error, but misorders the merging of taxon43 and taxon5. NeuralNJ-RL further improves the reconstruction, achieving the lowest RF distance (0.064) among all methods. Overall, RAxML-NG and Phyloformer exhibit substantial topological errors (RF distances of 0.234 and 0.277, respectively), while the NeuralNJ variants demonstrate progressively improved reconstruction accuracy, with RF distances decreasing from 0.128 (NeuralNJ) to 0.085 (NeuralNJ-MC) and 0.064 (NeuralNJ-RL).

[Figure 6](#) illustrates the dynamic tree construction processes of NeuralNJ and distance-based NJ methods. Using simulated multiple sequence alignment (MSA) data comprising 50 taxa and 1,024 nt generated under the GTR+I+G model (provided in [Supplementary Section F](#)), we examined an ideal case where NeuralNJ achieves correct topology prediction and NJ uses the true distance matrix from the reference tree.

The NJ algorithm demonstrates a region-focused construction pattern. In the initial steps (1–15), it merges deep leaf

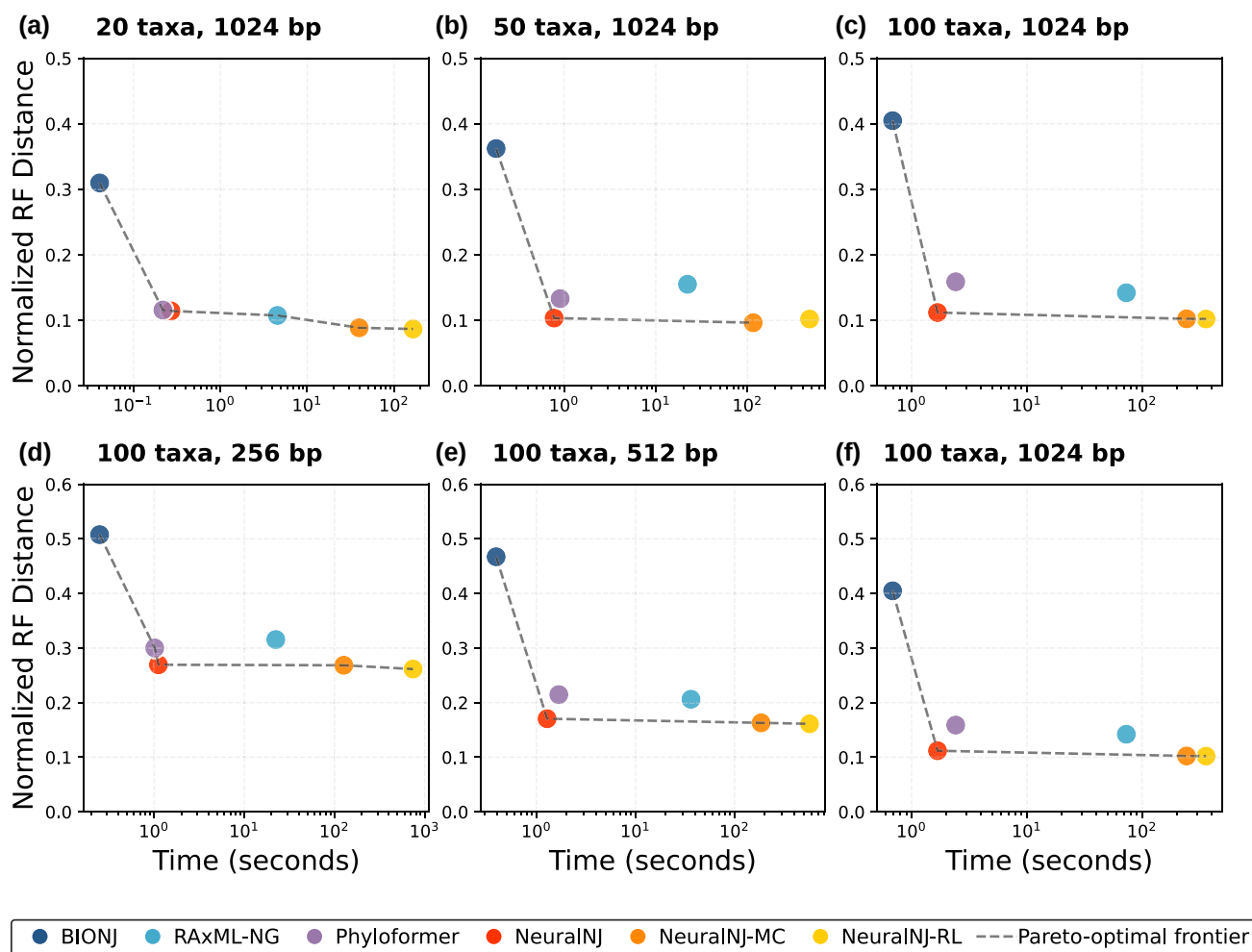


Fig. 4. Efficiency-accuracy trade-offs of phylogenetic inference methods under the GTR+I+G model. a)–c) Accuracy versus runtime across taxon counts with fixed sequence length (1024 nt). d)–f) Comparison across sequence lengths with fixed taxon number (100). The x-axis shows computational time in seconds (log scale), and the y-axis shows the normalized RF distance. Results are shown for six methods: BIONJ, RAXML-NG, Phyloformer, NeuralNJ, NeuralNJ-MC, and NeuralNJ-RL. The gray dashed line (Pareto frontier) represents the optimal trade-off between accuracy and computational time, where no method can improve one metric without compromising the other.

nodes such as taxon10 and taxon22, or small clusters like taxon21, taxon33, and taxon8. During steps 16–31, these small clusters are further joined with other taxa, including taxon1, taxon32–taxon19, and taxon16–taxon4, which all belong to the same major clade. This is followed by the progressive assembly of deeper internal nodes, ultimately forming the central topology of the tree. The Q-matrix calculation mechanism of NJ algorithm preferentially constructs branches that are close to each other while maintaining considerable distance from other sequences in the dataset. Although the NJ method guarantees accurate tree reconstruction when distance estimates are precise, this stepwise joining behavior may potentially lead to error accumulation in practical applications where distance estimates contain inaccuracies.

Our NeuralNJ method exhibits a distinctive construction pattern where merge operations are distributed across different regions of the tree simultaneously. As shown in the Fig. 6 (Steps 1–15 and 16–31), the algorithm concurrently processes multiple regions, merging taxon28 and taxon40 in the upper region, taxon34 and taxon39 in the middle, and taxon17 and taxon7 in the lower region. This distributed merging strategy differs fundamentally from NJ's region-focused approach. This behavior emerges from NeuralNJ's learning

objective: trained on labeled trees, it learns to identify and prioritize correct pairwise relationships globally, regardless of their regional location. Consequently, the algorithm achieves a more balanced tree construction process, potentially reducing the risk of error propagation that could occur from consecutive merges within a single region.

Discussion

This study presents NeuralNJ, demonstrating strong performance in phylogenetic inference by achieving an optimal balance between accuracy and efficiency. The end-to-end neural architecture enables direct topology learning and efficient single-pass tree construction, eliminating error accumulation inherent in iterative optimization approaches. Experiments on simulated datasets of varying scales demonstrate that NeuralNJ is robust and particularly excels in complex scenarios and large-scale problems with 100 species. On empirical datasets, NeuralNJ consistently shows competitive accuracy while significantly reducing computational overhead. Case studies further validate that our end-to-end distributed tree construction effectively minimizes error propagation, enhancing the overall accuracy of phylogenetic reconstruction.

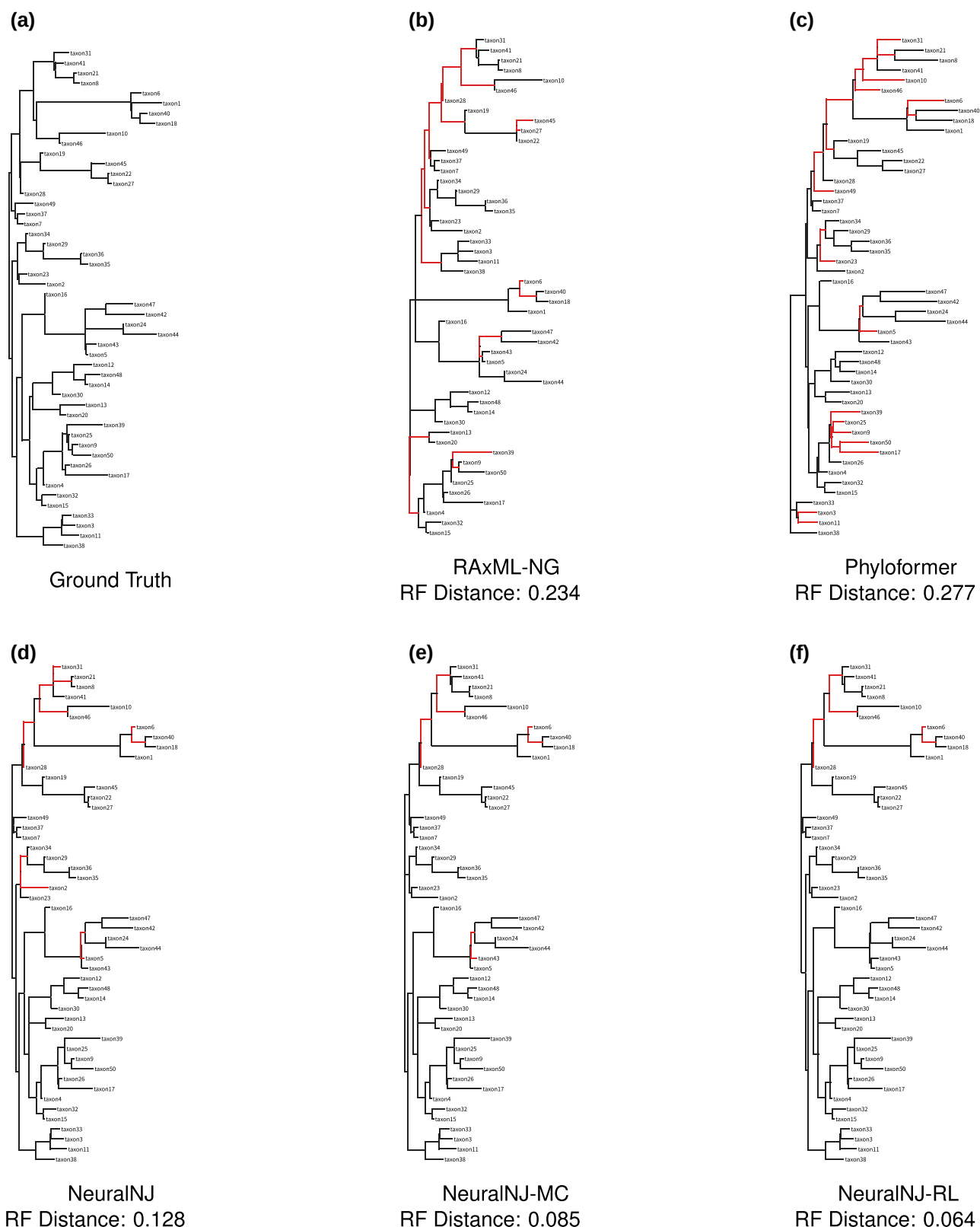


Fig. 5. Comparison of phylogenetic tree reconstructions across different inference methods on a simulated dataset (50 taxa, 1,024 nt) under the GTR+I +G model. a) Ground truth phylogenetic tree used for simulation. b)–f) Trees inferred by RAxML-NG, Phyloformer, NeuralNJ, NeuralNJ-MC, and NeuralNJ-RL, respectively. Topological discordance with the ground truth tree is highlighted (red branches), while conserved relationships are shown in black. The normalized RF distances between each inferred tree and the ground truth are shown. The sequence alignment used in this analysis is provided in [Supplementary Section E](#).

The exceptional performance of NeuralNJ stems from three key architectural innovations: an end-to-end neural framework for training and tree construction, **learning-to-rank**

strategy for join operations, and **topology-aware parent node representation estimation**. First, the end-to-end architecture enables direct optimization towards ground-truth tree

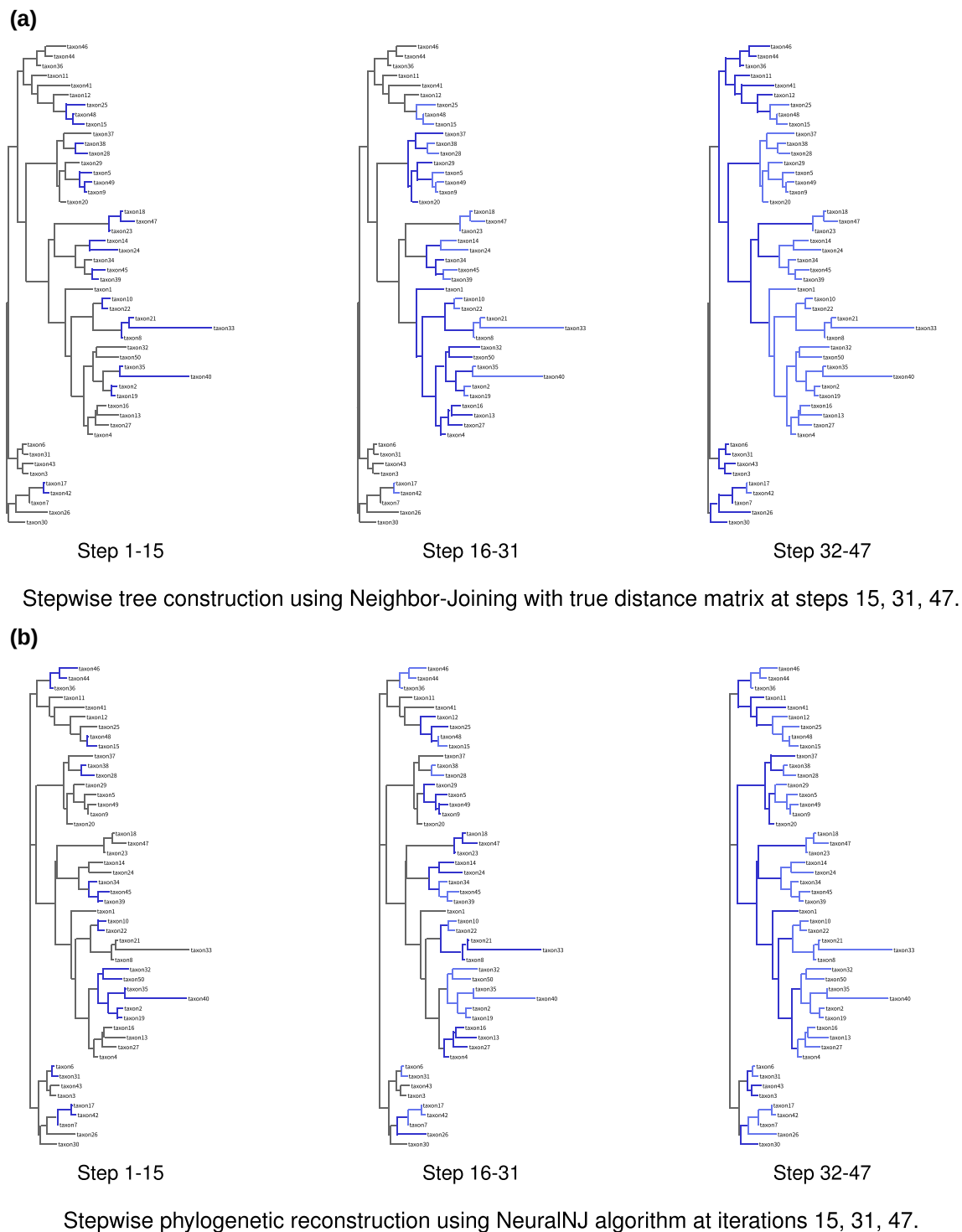


Fig. 6. Stepwise phylogenetic tree reconstruction comparing NJ and NeuralNJ on simulated sequence data (50 taxa, 1,024 nt) under the GTR+I+G model. a) Tree construction process using classical NJ with the true evolutionary distance matrix and b) NeuralNJ algorithm, shown at steps 15, 31, and 47. Branch colors denote merging order: black represents unmerged taxa, blue indicates the merging events that occurred in the current step being displayed, and light blue shows earlier merging events. Both methods converge to identical topologies (RF distance = 0). The sequence alignment used in this analysis is available in [Supplementary Section F](#).

topology reconstruction, while eliminating the time-consuming iterative topology optimization through single-pass inference. Second, learning join priorities instead of direct distance matrix estimation offers dual advantages: it simplifies the learning objective by focusing on correct pairing order rather than precise distance values, enhancing model scalability; and it allows the model to concentrate exclusively on high-priority pairs, making the learning process more focused and

efficient. Third, our topology-aware parent node representation incorporates not only immediate child node features but also innovatively integrates ancestral context, with ablation studies demonstrating significant performance improvements from this contextual integration.

While NeuralNJ effectively learns to map genome sequences to tree topologies from simulated data, its performance is inherently bounded by data quality and the similarity between

target problems and simulation data. We observe notable disparities in accuracy between simulated and empirical datasets, with empirical data showing consistently higher normalized RF distances. This gap likely stems from several factors in the current simulation-based training paradigm. First, the scarcity of real-world labeled trees prevents direct training on empirical data. Second, the GTR+I+G evolutionary model used in our simulations may not optimally fit all empirical datasets, despite evolutionary model selection being crucial for phylogenetic inference. Traditional phylogenetic inference typically involves optimal model selection, an aspect that remains unexplored in the context of neural networks and simulated data. Additionally, our current approach of using uniform tree topology distributions and exponential branch length distributions may not fully capture real-world evolutionary patterns, suggesting the need for more sophisticated simulation strategies aligned with empirical data distributions.

Despite the efficient computation of neural network methods for phylogenetic inference, their broader application remains constrained by hardware limitations. Due to GPU memory constraints, current neural network methods, such as NeuralNJ and Phyloformer, are typically limited to processing sequences shorter than 5,000 nucleotides for problems involving 100 taxa, and sequences under 17,000 nucleotides for datasets with 50 taxa. These technical limitations significantly restrict the application of neural approaches to large-scale phylogenomic datasets, which often contain concatenated alignments spanning hundreds of thousands to millions of sites. Future advancements in this field will necessitate the development of memory-efficient architectures capable of handling substantially longer sequence alignments while maintaining computational tractability.

NeuralNJ's flexible framework integrates both likelihood-free inference and likelihood-guided optimization within a unified approach. NeuralNJ directly infers phylogenetic tree topologies from sequence data, effectively circumventing the computationally expensive likelihood calculations that constrain ML methods when analyzing complex evolutionary models (Nesterenko et al. 2025). However, the base NeuralNJ model, trained on trees sampled from predefined evolutionary distributions, is fundamentally limited by its reliance on the training prior, which constrains its generalization to out-of-distribution data and may lead to suboptimal solutions for specific empirical cases. To address this limitation, NeuralNJ-MC and NeuralNJ-RL variants incorporate likelihood-based tree selection and reinforcement learning optimization, enabling the model to adapt its search beyond the constraints of the training prior and improve performance on individual instances. Experimental results indicate that calibrated likelihood guidance generally enables models to explore broader tree space and can optimize both likelihood and topological accuracy simultaneously. However, representative cases presented in the Supplementary Material show that likelihood optimization does not always guarantee improved RF distance, and the relationship between likelihood and topology remains complex and case-dependent. These results underscore the complexity of the relationship between likelihood and topology, indicating promising directions for future research.

While NeuralNJ-MC and NeuralNJ-RL demonstrate superior accuracy compared to the base NeuralNJ model, this improvement comes at the cost of intensive likelihood calculations and branch length optimizations. Notably, the

computational demands of NeuralNJ-RL's fine-tuning process are particularly significant, requiring substantial GPU resources for each case, which emphasizes the necessity for more efficient search strategies. Future research directions could productively explore neural approaches for direct branch length prediction and efficient likelihood estimation, while simultaneously investigating methods to reduce the impact of phylogenetic prior biases in the training data.

Our work on NeuralNJ, together with its variants and recent developments in deep learning approaches for phylogenetic inference, represents a significant step towards efficient and accurate evolutionary tree reconstruction. The demonstrated balance between computational efficiency and accuracy, particularly in complex evolutionary scenarios, will contribute to better understanding of evolutionary relationships across diverse species. Future developments in simulation data quality and neural network architectures will further enhance the applicability of deep learning methods in phylogenetic analysis, potentially revolutionizing our approach to studying evolutionary history.

Materials and Methods

Data and Implementation

Simulated phylogenetic data. We generated simulated phylogenetic data using a three-stage approach informed by empirical observations. First, tree topologies were generated using uniform random binary trees via Biopython (Cock et al. 2009; Talevich et al. 2012). Next, branch lengths were sampled from exponential distributions with rate parameters λ (Rannala and Yang 1996; Stadler 2011) derived from real phylogenetic datasets (Azouri et al. 2021). Based on the analysis of these empirical data, λ values were drawn uniformly from $[\log 2, \log 5]$ in log space. DNA sequences were simulated with IQ-TREE2's Alisim (Ly-Trong et al. 2022, 2023) under the GTR+I+G model (Tavaré 1986). For each alignment, substitution rates, base frequencies, gamma shape, and invariant site proportions were sampled from empirical distributions (Naser-Khdour et al. 2021). To reproduce realistic gap patterns, deletion rates were varied within $[0, 0.01, 0.02, 0.03, 0.04]$, and rates below 0.04 were chosen to keep simulated gap ratios below 0.3, consistent with empirical observations (10–30%) (Zhou et al. 2017).

The training dataset consists of 29,000 unique tree-MSA pairs, with each phylogenetic tree simulated independently and paired with exactly one corresponding MSA. All training samples contain 50 taxa, while sequence lengths vary from 128 to 1,024 nucleotides in increments of 32 (resulting in 29 different sequence lengths, with 1,000 unique tree-MSA pairs per length). For the test dataset, we similarly generated unique tree-MSA pairs (each consisting of one independently simulated phylogenetic tree and its corresponding MSA) according to the following design: 128 pairs for each combination of sequence length 1,024 and taxa counts of 20, 50, and 100; additionally, for the 100-taxa case, we included 128 pairs each for sequence lengths of 256 and 512 nucleotides.

Evolutionary models. We employed the GTR+I+G model for our main analyses, with all model parameters (substitution rates, base frequencies, gamma shape, and invariant site proportion) sampled from empirical distributions (Naser-Khdour et al. 2021). This approach allows for varying substitution rates among nucleotides, rate heterogeneity across sites (+G), and invariable sites (+I), making it particularly suitable for real-world phylogenetic analyses (Sumner et al. 2012;

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Arenas 2015). Additional analyses incorporating using the simpler Jukes-Cantor (JC) model (Jukes and Cantor 1969) are presented in the [Supplementary Materials](#).

Empirical datasets. We evaluated NeuralNJ on empirical phylogenomic data from Zhou et al. (2017), which included four distinct DNA collections: WickD3b (land plants), JarvD5a (birds), SongD1 (mammals), and TarvD7 (mammals). For datasets with fewer than 50 taxa (JarvD5a, SongD1, TarvD7), we included sequences up to 17,000 nucleotides; for WickD3b (up to 100 taxa), we processed sequences up to 4,400 nucleotides. This approach achieved thorough dataset coverage: SongD1 (100%), JarvD5a (98%), TarvD7 (99.8%), and WickD3b (50%, limited by available reference trees). For each dataset, we used the reference tree topology inferred by ML estimation on concatenated genome sequences (Nesterenko et al. 2025).

Baselines for comparison. We selected a range of baseline methods for phylogenetic inference, including the heuristic distance-based algorithm BIONJ (Gascuel 1997), the well-established ML phylogenetic solver RAXML-NG (Kozlov et al. 2019), and the neural network-based inference algorithm Phyloformer (Nesterenko et al. 2025). BIONJ represents a commonly used heuristic approach for initial tree generation in evolutionary solvers. RAXML-NG represents a highly optimized, mature phylogenetic solver. Phyloformer, a recent development, exemplifies state-of-the-art neural network approaches in this field. By comparing our proposed method with these established techniques, we aimed to comprehensively evaluate its performance and effectiveness in phylogenetic inference. It is worth noting that both NeuralNJ and Phyloformer were trained on identical training data and evaluated using a consistent early stopping mechanism (training terminates when validation performance shows no improvement for five consecutive validation checks) to ensure fair comparison.

Evaluation metric. We primarily evaluated NeuralNJ using the Normalized RF distance (Robinson and Foulds 1981), which measures the structural differences between inferred and reference trees. This metric directly quantifies the topological accuracy of phylogenetic inference methods by comparing the shared splits between two trees, with values ranging from 0 (identical topologies) to 1 (completely different topologies). Additional analyses, including likelihood-based evaluations under different branch length optimization strategies, are presented in the [Supplementary Materials](#).

Model Design of NeuralNJ

The proposed NeuralNJ framework comprises three essential neural modules for phylogenetic tree construction: (1) a sequence encoder ($\mathcal{M}_{\text{enc}}$) that transforms MSA input into learned sequence representations, and a tree decoder consisting of (2) a parent node representation estimator ($\mathcal{M}_{\text{est}}$) that conducts topology-aware representation estimation of potential ancestral nodes for candidate subtree pairs, and (3) a subtree evaluator ($\mathcal{M}_{\text{eval}}$) that determines joining priorities based on these ancestral representations. The architectural details of these modules are elaborated in the following sections, while their complete mathematical formulations are presented in [Supplementary Section C](#).

(1) **Sequence encoder $\mathcal{M}_{\text{enc}}$.** The sequence encoder $\mathcal{M}_{\text{enc}}$ in NeuralNJ adopts the axial attention architecture from MSA-Transformer (Rao et al. 2021), which effectively

captures evolutionary relationships between species while accommodating variable sequence lengths and quantities.

The encoding process consists of three main steps. First, for an input MSA $X = \{x_i\}_{i=1}^N$ containing N sequences, each sequence x_i of length L is embedded into a high-dimensional space through a multilayer perceptron (MLP) with a GELU-activated hidden layer, yielding embeddings $\tilde{x}_i = [\tilde{x}_{ij}]_{j=1}^L$.

Second, these embeddings undergo transformation through a cascade of axial attention layers (Ho et al. 2019), following the MSA-Transformer architecture (Rao et al. 2021). Each layer implements two complementary attention mechanisms: (i) column attention, which computes position-wise correlations between species embeddings, and (ii) tied row attention, which employs a shared attention map across species to capture positional correlations within individual sequences. Both mechanisms operate with a memory complexity of $O(LN^2 + L^2)$.

Finally, the encoder produces $H = \{h_i\}_{i=1}^N$ for the N species, where each species' representation h_i is a composite of its sequence embedding representations after the attention transformations, i.e. $h_i = [h_{ij}]_{j=1}^L$. This encoding process facilitates the transfer of information across species sequences, effectively capturing inter-species evolutionary information.

(2) **Parent node representation estimator $\mathcal{M}_{\text{est}}$.** As illustrated in Fig. 1c, the parent node representation estimator $\mathcal{M}_{\text{est}}$ computes the parent node representation h_x for two subtrees with representations h_l and h_r that are to be joined. In phylogenetic tree construction, the parent node serves as a bridge between its immediate descendants and more distant ancestral nodes. This dual role imposes two key requirements: First, h_x must maintain strong correlations with both h_l and h_r , as it represents their immediate common ancestor. Second, h_x should also capture relationships with potential ancestral and sibling nodes, whose information is embedded within the broader subtree set excluding h_l and h_r .

To achieve topology-aware parent node representation estimation that satisfies these requirements, $\mathcal{M}_{\text{est}}$ computes h_x through a two-stage process through hierarchical aggregation:

Stage 1: immediate descendant integration. A lightweight gating network performs site-by-site aggregation of the subtree representations $h_l = [h_{lj}]_{j=1}^L$ and $h_r = [h_{rj}]_{j=1}^L$ to generate an intermediate representation $\tilde{h}_x$:

$$\begin{aligned} z_j &= \text{sigmoid}(\text{Linear}(h_{lj} - h_{rj})), \\ \tilde{h}_{xj} &= z_j \odot h_{lj} + (1 - z_j) \odot h_{rj}, \end{aligned} \quad (1)$$

where j denotes the site index, and the gating mechanism adaptively weights the contribution of each subtree's features.

Stage 2: ancestral context integration. The final representation h_x is computed by integrating $\tilde{h}_x$ with a context-aware representation h_{rest} through another site-specific gating network:

$$\begin{aligned} w_j &= \text{sigmoid}(\text{Linear}(h_{\text{rest}j})), \\ h_{xj} &= w_j \odot \tilde{h}_{xj} + (1 - w_j) \odot h_{\text{rest}j}. \end{aligned} \quad (2)$$

The context-aware representation h_{rest} encapsulates information from potential ancestral and sibling nodes by aggregating the complementary subtree set (excluding h_l and h_r). This aggregation is implemented through a cross-attention mechanism (Vaswani et al. 2017), using $\tilde{h}_x$ as the query and the remaining subtree representations $\{h_k, k \neq l, r\}$ as keys:

$$h_{\text{rest}} = \text{CrossAttention}(\tilde{h}_x, \{h_k, k \neq l, r\}). \quad (3)$$

Notably, while the gating networks operate site-wise to preserve fine-grained sequence features, the cross-attention mechanism processes complete sequence representations to maintain comprehensive evolutionary information in h_{rest} . The detailed computational procedure is provided in [Algorithm 3 in Supplementary Section C](#).

Through this hierarchical two-stage process, the representation aggregator efficiently computes the parent node representation $h_x = [h_{xj}]_{j=1}^L$. The combination of site-wise gating operations and cross-attention mechanism ensures that h_x effectively captures both local relationships with direct descendants and broader evolutionary context from potential ancestors and siblings. Importantly, the adaptive gating design maintains gradient stability during backpropagation, allowing robust optimization even for deep phylogenetic trees with large species sets. As demonstrated in [Supplementary Fig. S18](#), this two-stage aggregation process with attention-based ancestral context integration achieves superior convergence compared to simpler alternatives like child-only aggregation or sum-based integration. This topology-aware representation lays the foundation for subsequent priority prediction and tree construction.

(3) **Subtree Evaluator** $\mathcal{M}_{\text{Eval}}$. The subtree evaluator maps the aggregated node representation h_x to a scalar priority score s_{τ_x} for the newly-joined subtree τ_x . Specifically, a MLP first transforms each site-specific feature of $h_x = [h_{xj}]_{j=1}^L$ into a scalar value, and these values are then averaged across all sites to produce the final priority score. This site-wise averaging strategy not only ensures computational efficiency but also provides balanced consideration of evolutionary signals from all sites.

Subtree Selection Strategies

NeuralNJ's flexible framework supports multiple strategies for selecting subtree pairs based on the predicted priority scores s_{τ_x} . NeuralNJ employs a direct greedy selection strategy, while NeuralNJ-MC and NeuralNJ-RL utilize probabilistic selection mechanisms to enable likelihood-guided search.

The greedy strategy, implemented in NeuralNJ, directly selects the newly-joined subtree with the highest priority score:

$$\tau_x^* = \underset{\tau_x \in \mathcal{A}_t}{\operatorname{argmax}} s_{\tau_x}, \quad (4)$$

where $\mathcal{A}_t$ represents the set of all candidate newly-joined subtree at step t .

For enhanced topology exploration in NeuralNJ-MC and NeuralNJ-RL, we implement probabilistic selection strategies. These approaches convert priority scores into sampling probabilities using a temperature-scaled softmax function. For notational convenience, we express this probability distribution as:

$$\pi_{\theta}(a_t | S_t) = \operatorname{softmax}\left(\left[\frac{s_{\tau_x}}{\gamma}, a_t = \tau_x \in \mathcal{A}_t\right]\right), \quad (5)$$

where S_t denotes the subtree set at step t , γ is the sampling temperature that controls exploration breadth. This probabilistic framework serves as the foundation for likelihood-guided search through Monte-Carlo sampling and reinforcement learning, respectively.

Supervised Training and Reinforced Search

Supervised training. To train the NeuralNJ algorithm supervisedly using the existing simulated MSA phylogenetic tree

dataset, we design a training algorithm based on NeuralNJ's bottom-up tree construction process. Given an MSA and its corresponding ground-truth tree τ_{gt} , we reconstruct τ_{gt} with NeuralNJ and optimize the priority scores at each construction step.

At the t th step, we identify two sets of candidate newly-joined subtrees: those that are part of τ_{gt} and those that are not. To ensure accurate reconstruction, candidates belonging to τ_{gt} should receive higher priority scores. We formalize this intuition using a binary margin-based pairwise ranking loss, which adopts the hinge loss formulation to enforce a margin between positive (ground-truth) and negative subtree candidates ([Burgess et al. 2005](#)):

$$\mathcal{L}_t = -\frac{1}{|\mathcal{P}_t|} \sum_{(\tau_a, \tau_b) \in \mathcal{P}_t} \max(s_{\tau_a} - s_{\tau_b} - m, 0), \quad (6)$$

where $\mathcal{P}_t$ denotes all pairs (τ_a, τ_b) with $\tau_a \in \mathcal{A}_t$ being a correct subtree (part of τ_{gt}) and $\tau_b \in \mathcal{A}_t$ being an incorrect one at step t , s_{τ} represents the priority score for subtree τ , and m is the margin hyperparameter.

This ranking loss is theoretically motivated by the RF distance, which quantifies topological differences between two phylogenetic trees by counting the number of unique splits present in one tree but not in the other ([Robinson and Foulds 1981](#)). By encouraging higher priority scores for correct splits while penalizing incompatible ones at each step, our loss function effectively guides the tree construction process to maximize the number of shared splits between the reconstructed tree and the ground truth tree, thereby minimizing their RF distance.

This loss function design directly supports our greedy selection strategy. By requiring all correct subtrees to have scores at least m higher than all incorrect subtrees, when this margin condition is satisfied across all possible pairs, the highest-scoring subtree will necessarily be from the set of correct subtrees. This formulation allows the model to focus on difficult cases where the margin is insufficient while ignoring already well-predicted pairs, improving training stability and effectiveness. The margin-based approach effectively teaches the neural network to prioritize correct joining operations during inference, making our greedy selection strategy both efficient and accurate.

Notably, among the candidate newly-joined subtrees $\mathcal{A}_t$, incorrect candidates typically constitute a large majority. We therefore adopt a selective penalization strategy, focusing on incorrect candidates with higher priority scores. This approach serves two purposes: it enhances the model's accuracy in identifying high-priority subtrees, thereby improving overall tree construction accuracy, while also reducing the cardinality of $\mathcal{P}_t$, leading to optimized computational overhead and increased training efficiency. The detailed selection techniques are elaborated in the Implementation Details section of the [Supplementary Material](#).

The overall supervised learning objective is obtained by averaging the ranking losses across both the batch samples and tree construction steps:

$$\mathcal{L}_{\text{SL}} = \frac{1}{M} \sum_{i=1}^M \frac{1}{N-2} \sum_{t=1}^{N-2} \mathcal{L}_t^{(i)}, \quad (7)$$

where M denotes the batch size, N is the number of species, and $N-2$ represents the total number of joining operations required to construct a complete binary tree.

Reinforced search. NeuralNJ-RL leverages reinforcement learning to enhance its tree construction strategy for specific phylogenetic inference tasks, naturally aligning with the sequential decision-making nature of tree construction. While our supervised learning phase minimizes the RF distance to simulated ground truth trees, this approach may have limited generalization capability when encountering evolutionary patterns that deviate from the simulation assumptions. To enhance model adaptability to the specific case we employ phylogenetic likelihood $\log P(X|\tau)$ (Felsenstein 1981) as the reward signal, directly optimizing this well-established measure that captures the fit between the tree topology and the specific observed sequence data. By adopting the REINFORCE policy gradient algorithm (Williams 1992) with the probabilistic selection mechanism defined in Equation 5, it effectively balances exploration of diverse tree topologies with exploitation of promising construction paths. Its learning objective is defined as:

$$\mathcal{L}_{\text{RL}} = -\frac{1}{M} \sum_{i=1}^M \log p(\tau^{(i)} | \theta) (R(\tau^{(i)}) - b), \quad (8)$$

where $p(\tau^{(i)} | \theta) = \prod_{t=1}^N \pi_{\theta}(a_t^{(i)} | S_t^{(i)})$.

Here, M denotes the batch size, $p(\tau^{(i)} | \theta)$ represents the likelihood of constructing the i th sampled tree under parameters θ . $R(\tau^{(i)}) = \log P(X|\tau^{(i)})$ is the reward, computed as the phylogenetic likelihood of the final topology, and b is a baseline reward estimated from the previous epoch to reduce gradient variance. By maximizing this objective, the model learns to assign higher probabilities to construction paths that lead to trees with greater phylogenetic likelihood. This likelihood-guided learning enables the model to capture the long-term consequences of its joining decisions in the vast space of possible phylogenies.

Supplementary Material

Supplementary material is available at *Molecular Biology and Evolution* online.

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Conflict of interest

None declared.

Data Availability

All data, code, and analysis files associated with this study are freely available at <https://github.com/ZhangXinru99/NeuralNJ>.

Appendix

Technical Primer for NeuralNJ

Transformer architectures. The transformer is a neural network design that excels at processing sequential data (like DNA/protein sequences) by letting all elements communicate directly through attention mechanisms including self-attention and cross-attention. Unlike traditional neural networks such as Recurrent Neural Networks (Schuster and Paliwal 1997) that process sequences word-by-word, or Convolutional Neural Networks (Gu et al. 2018) that analyze local patterns through fixed window sizes, transformers analyze all positions simultaneously through information exchange. The core architecture contains stacked processing units that progressively refine sequence understanding through two complementary components: (1) An encoder that integrates contextual information across the entire sequence, and (2) A decoder that generates predictions based on these learned patterns. This parallel processing enables efficient modeling of long-range evolutionary dependencies that span entire genomes or protein families. The power of this architecture is exemplified by breakthroughs like AlphaFold2 (Jumper et al. 2021), where transformer-based processing of multiple sequence alignments (MSAs) enables accurate protein structure prediction by capturing co-evolutionary patterns across distant sequence positions.

Cross-attention mechanism. Cross-attention facilitates adaptive information integration between biological entities through query-key-value interactions. The mathematical formulation is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

with distinct biological interpretations: Queries (Q) originate from the target representation (e.g. ancestral nodes being estimated in NeuralNJ), and Keys (K) and Values (V) derive from the source representation (e.g. sibling subtrees in NeuralNJ).

Self-attention mechanism. The self-attention mechanism computes contextual relationships within a single biological sequence through learnable interactions between all position pairs. It calculates attention using the same mathematical foundation as cross-attention, but with Q, K, and V all derived from the same input sequence. This allows each nucleotide/amino acid position to dynamically weight its evolutionary dependencies on other sites.

Axial-attention mechanism. For multiple biological sequences, we often use axial attention, a modified self-attention that separately handle row-wise (within-species) and column-wise (across-species) relationships in sequence alignments, imitating how evolutionary patterns form through both vertical inheritance and horizontal constraints. The axial-attention approach has demonstrated success in key biological applications including protein contact prediction, mutation effect forecasting, and evolutionary coupling analysis (Rao et al. 2021), making it particularly suitable for phylogenetic tasks requiring simultaneous analysis of within-species variation and cross-species conservation patterns.

Gated mechanism. Gating is a neural network technique that dynamically regulates information flow using learnable filters. These adaptive gates employ mathematical functions (like sigmoid) to compute precise blending ratios (0–1) between different inputs at each sequence position to preserve critical features while suppressing noise. In NeuralNJ, this

enables selective integration of evolutionary signals during ancestral node estimation, where gates automatically determine which descendant mutations to retain and which ancestral patterns to emphasize. Similar gating strategies have proven essential in cutting-edge biological models like AlphaFold2, where they help balance local structural constraints with global folding patterns during protein structure prediction.

MLP. The MLP operates as a universal function approximator through hierarchical nonlinear transformations. Each layer performs affine projection $h^{(l)} = W^{(l)}x^{(l-1)} + b^{(l)}$ followed by ReLU activation $\max(0, h^{(l)})$, progressively distilling high-dimensional embeddings into compact evolutionary signals.

NeuralNJ Architecture for Phylogenetic Inference

NeuralNJ employs a transformer-based encoder-decoder architecture specifically designed for phylogenetic tree construction. The encoder processes multiple sequence alignments (MSAs) using axial-attention mechanisms which can independently analyze vertical relationships (comparing homologous sites across species columns) and horizontal patterns (examining sequence context within individual rows). This dual-axis approach generates evolutionary embeddings that simultaneously encode cross-species conservation and lineage-specific variations.

The decoder translates these representations into tree-building decisions through a hierarchical process. Its core component, the parent estimator, operates in two phases: (1) immediate descendant integration through a lightweight gating network that combines direct child node features, and (2) sibling node aggregation using cross-attention mechanisms and gating network to integrate complementary subtree representations (excluding direct descendants). This two-stage aggregation with ancestral context integration achieves superior topological reconstruction while maintaining biologically plausible evolutionary relationships.

The subtree evaluator transforms the aggregated node representation into scalar priority scores for new subtree configurations. Specifically, a MLP processes the parent node's estimated representation to generate final priority scores, which guide the selection of optimal subtree pairs during the tree-building process.

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